Journal of Allbiosolution
(2021) Volume 3, Issue 1, Article ID: A-0301014

Title: Bioactive compounds of *Allium sativum* (Garlic) & their pharmacological properties

Authors: Alok Tripathi[1], Surabhi Singhal[1], Navneet Sharma[1]*

aloktri08@gmail.com; singhalsurbhisur@gmail.com; navneet_admn@yahoo.co.in

[1]School of Life Science & Technology, IIMT University, Meerut, U.P., India

*Corresponding Author

Corresponding Author details

Dr. Navneet Sharma, Professor

School of Life Science & Technology,

IIMT University, Meerut

E-mail: navneet_admn@yahoo.co.in

Contact no.: 9412207971

ABSTRACT

Allium sativum (Garlic) is considered as most significant herbs that have been used as traditional medicine from very old times. From ancient to current time different studies have confirmed the significant biological activities of garlic which includes anti-infectious properties, anti-inflammatory, antioxidant, anti-obesity, cardiovascular protective, anti-diabetic, free radical scavenging and anticancer properties.

Allium sativum has a diverse range of bioactive compounds and these are mainly organosulfur compounds, phenolic compounds and saponins. Primarily dominating phytochemicals are sulfur-containing compounds which are Alliin ($C_6H_{11}NO_3S$), Allicin ($C_6H_{10}OS_2$), E-Ajoene ($C_9H_{14}OS_3$), Z-Ajoene ($C_9H_{14}OS_3$), 2-Vinyl-4H-1,3-dithiin ($C_6H_8S_2$), Diallyl sulfide ($C_6H_{10}S$), Diallyl disulfide ($C_6H_{10}S_2$), Diallyl trisulfide ($C_6H_{10}S_3$), Allyl methyl sulfide (C_4H_8S). It is found that bioavailability of capsule prepared by garlic powder is in the range of 26-109 % while bioavailability of non-enteric tablets is in the range of 80-111 %. It shows that bioavailability of tablets are more in comparison to capsule of garlic powder

Current review discusses the chemical bioactive compound compositions of the *Allium sativum* (Garlic) and its pharmacological properties, which prove the importance of garlic in the cure of different types of diseases.

Keywords: Garlic, *Allium Sativum*, Antioxidant, Anti-inflammatory, Anticancer

INTRODUCTION

Allium sativum (Garlic) is considered as kitchen spices which are used in the flavoring of food due to its so many valuable properties. It is native to central Asia but nowadays is cultivated in the whole world (Rasul *et al*., 2015). It is considered as most significant herbs that have been used as traditional medicine from very old times. There is a longer history of utilization of garlic, as it has therapeutic and prophylactic activities, due to which it is used as an ingredient in the different medicinal preparations (Singh *et al*., 2007; Ali *et al*., 2012). Ancient peoples utilized garlic as raw as well as in preparations for the curing of constipation, fever and different types of microbial infections. From ancient to current time different

studies have confirmed the significant biological activities of garlic which includes anti-infectious properties, anti-inflammatory, antioxidant, anti-obesity, cardiovascular protective, anti-diabetic, free radical scavenging and anticancer properties (Khanum *et al*., 2004; Borek *et al*., 2006; Lee *et al*., 2012, 2016; K. Charu *et al*., 2014; Bradley *et al*., 2016). These marvelous biological properties are proved by the different research results which are based on the clinical & epidemiological evidences (Guercioe *et al*., 2014).



Fig.1: Representation of the biological activities of the *Allium sativum* (Garlic)

TAXONOMIC CLASSIFICATION OF GARLIC

Garlic belongs to the family of onion i.e. Liliaceae. The taxonomic classification of garlic is as follows

Kingdom: Plantae

Division: Magnoliophyta

Class: Liliopsida

Subclass: Liliidae

Order: Liliales

Family: Liliaceae

Genus: *Allium*

Species: *sativum*

BIOAVAILABILITY OF GARLIC

It is found that bioavailability of major organo-sulfur containing components of Garlic greatly varies in supplement preparation and raw garlic. Majority of the pharmacological activity of raw garlic is due to the presence of compound Allicin. It is found that allicin is metabolized very quickly into allyl mercaptan within less than 1 minute outside of the body however; allicin as well as allyl mercaptan was not found in body even after consumption of a higher amount of raw garlic (Lawson *et al*., 1992). The bioavailability of garlic bioactive compounds varies in the form of they are prepared. It's found that bioavailability of capsule prepared by garlic powder is in the range of 26-109 % while bioavailability of non-enteric tablets is in the range of 80-111 %. It shows that bioavailability of tablets are more in comparison to capsules of garlic powder (Lawson *et al*., 2018).

CHEMICAL COMPOSITION OF *ALLIUM SATIVUM*

Allium sativum has a diverse range of bioactive compounds and these are mainly organosulfur compounds, phenolic compounds and saponins. Primarily dominating phytochemicals are sulfur-containing compounds which are Alliin ($C_6H_{11}NO_3S$), Allicin ($C_6H_{10}OS_2$), E-Ajoene ($C_9H_{14}OS_3$), Z-Ajoene ($C_9H_{14}OS_3$), 2-Vinyl-4H-1,3-dithiin ($C_6H_8S_2$), Diallyl

sulfide ($C_6H_{10}S$), Diallyl disulfide ($C_6H_{10}S_2$), Diallyl trisulfide ($C_6H_{10}S_3$), Allyl methyl sulfide (C_4H_8S) (Al-Snafi, 2013).

S. No.	Compounds	Molecular formula
1.	Allicin	$C_6H_{10}OS_2$
2.	Alliin	$C_6H_{11}NO_3S$
3.	E-Ajoene & Z-Ajoene	$C_9H_{14}OS_3$
4.	2-Vinyl-4H-1,3-dithiin	$C_6H_8S_2$
5.	Diallyl sulfide (DAS)	$C_6H_{10}S$
6.	Diallyl disulfide (DADS)	$C_6H_{10}S_2$
7.	Diallyl trisulfide (DATS)	$C_6H_{10}S_3$
8.	Allyl methyl sulfide (AMS)	C_4H_8S

Table.1: Chemical components of garlic and their molecular formula

Most important Cysteine sulfoxide of the garlic is Alliin, which is getting converted into another cysteine sulfoxide allicin. This conversion is induced by cutting of garlic in presence of enzyme alliinase. Specific intense odor of the garlic is due to the presence of the component such as allicin, S-methyl-cysteine-sulfoxide and S-propyl-cysteine-sulfoxide (Zeng *et al*., 2017). These cysteine sulfoxides have the ability to produce a number of arrays of metabolites depending upon different physiological conditions. For example S-propyl-cysteine-sulfoxide provides approximately fifty metabolites by varying the condition of temperature, water content and alliinase enzyme. These can produce different products such as methyl-methane-thio-sulfonate and thio-sulfinates from the reactant mixture of S-methyl-cysteine-sulfoxide, S-propyl-cysteine-sulfoxide and alliin (Zeng *et al*., 2017).

The organosulfur compounds of garlic are, may be, of two different categories such as water soluble organosulfur compounds and oil soluble organosulfur compounds. Water soluble organosulfur compounds such as allyl-mercaptan (AM), allyl-methyl-sulfide (AMS), S-allyl-cysteine (SAC) and S-allyl-mercapto-cysteine (SAMC) have important role in the cancer prevention i.e. anticancer bioactive component. However, the amount of water soluble organosulfur compounds is very low in comparison to oil soluble organosulfur compounds (Fukushima *et al*., 2001).

PHARMACOLOGICAL PROPERTIES OF GARLIC

Antimicrobial properties

Garlic has varying degrees of antimicrobial properties against different microorganisms such as bacteria, fungi and protozoans. On the basis of affectivity against different microbes

antimicrobial properties could be categorized as antibacterial, antifungal & anti protozoans. Extract prepared by the raw garlic may be aqueous as well as alcoholic and both represent varying levels of vulnerability against the disease causing microbes.

Antibacterial activities

Garlic extract shows antimicrobial activity against a wide range of bacteria which may be Gram-positive, Gram-negative or antibiotic-resistant. Wide range of antibacterial activity has been recorded by different studies against *Escherichia coli*, *Klebsiella* sp., *Mycobacterium*, *Pseudomonas aeruginosa*, *Salmonella* sp., *Shigella* sp., *Staphylococcus aureus*, *Salmonella* sp. and *Vibrio* sp. (Ross *et al.*, 2001; Kuda *et al*., 2004; Cutler & Wilson, 2004; Wallock-Richards *et al*., 2014). This antibacterial activity of the garlic extract is attributed by bioactive compound Allicin presence in it.

Antifungal activities

Garlic has a wide range of antifungal properties as shown by this extract against different fungal species such as *Aspergillus*, *Candida*, *Meyerozyma guilliermondii*, *Rhodotorula mucilaginosa*, *Trichophyton* and *Trichosporon*. On the basis of effectiveness results it could be elaborated that garlic extract shows broad spectrum antifungal effect. Aqueous as well as ethanolic extract of garlic represent antifungal effect against *Aspergillus flavus*, *A. niger*, *Botrytis cinerea*, *Candida* sp., *Penicillium expansum* and *Rhizopus stolonifer* (Fufa, 2019; Pârvu *et al*., 2019).

Antiprotozoal activities

There are several diseases caused by the protozoans which are life threatening if they are not cured timely by specific antiprotozoal properties. According to several research results it is now proved that garlic extract has potential to retard the growth of protozoan & help in combating the diseases caused by protozoans. Garlic extract has several bioactive compounds in which Ajoene show anti-protozoan activity against *Trypanosoma cruzi* responsible for chagas diseases and this activity is due to the targeting of trypanothione reductase enzyme of *Trypanosoma cruzi* by the Ajoene (Gallwitz *et al*., 1999). One of the researches carried out *in-vivo* on *Blastocyst, a* protozoan, to demonstrate the antiprotozoal effect of garlic extract and found that garlic extract has the ability to inhibit the growth of *Blastocyst*. This antiprotozoal activity is due to the thiosulfinate compound, a bioactive compound present in garlic extract (Abdel-Hafeez *et al*, 2015). However, research carried out by the researcher Gruhlke, shows that garlic extract has anti-protozoan effects against wide range of parasitic protozoans such as *Entamoeba histolytica*, *Giardia species*, *Plasmodium falciparum* and *Trypanosoma species*. According to him, this activity is attributed to one of the most important bioactive compounds of garlic extract allicin (Gruhlke *et al*., 2016).

Antioxidant activity

Antioxidant activity of any organism is dependent on the level of important antioxidant enzymes of the cells mainly glutathione peroxidase, superoxide dismutase and catalase. It is found in the research that antioxidant activity of the garlic extract is far better than raw garlic and the reason behind this is garlic extract has more organosulfur compounds in comparison to raw garlic (Ichikawa *et al*., 2006). It is found that garlic extract has the properties to uplift the level of antioxidant enzyme of the cells due to the presence of different phytochemicals in it. Increased level of the antioxidant enzymes represents the good level of defense system of the body (Liu *et al*., 1992; Borek, 1997). Garlic has potential to modulate the expression of different liver functioning enzymes such as SGOT, SGPT & ALP, which has an important role in the detoxification of toxic substances by removing the oxidative stress of the cell. This activity of garlic is due to the presence of organo-sulphur compounds, which proves it as a good antioxidant substance (Ide & Lau, 2001).

One of the researches on RAW264.7 macrophages shows that garlic has immunomodulatory properties which may modulate the expression of different inflammatory cytokines such as interferon- γ, IL-1, IL-10, IL-6 and TNF-α. These inflammatory cytokines expression ultimately leads to inflammation & cancer genesis (Li *et al*., 2017; Liu *et al*., 2017).

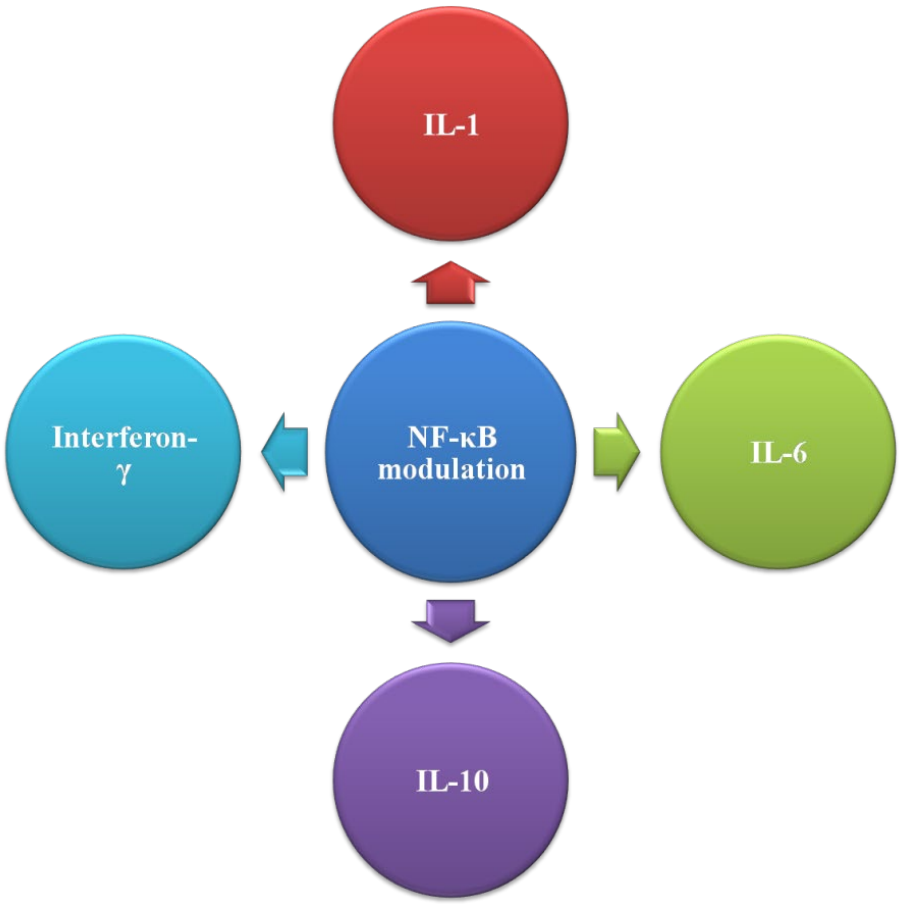


Fig.2: Representation of NF-□B modulation on inflammatory cytokines

Along with antioxidant properties garlic shows other protective effects which leads to decrease of cellular toxicity and effect is decline of reactive oxygen species i.e. ROS. The reason behind this is synergistic interaction of different biological ingredients like sulphur rich compounds, saponins and flavonoids (Borek, 2001).

Anti-Inflammatory activity

Garlic extract has ability to show anti-inflammatory effect by inhibiting the expression of transcription factor NF-□B. It is an important transcription factor which plays a central role in the inflammation by the modulation of the expression of different cytokine expression. Unregulated expression of transcription factor NF-□B represents the disease condition and responsible for many immune system related diseases such as cancer and autoimmune disease etc (Tripathi *et al*., 2019). However, it's proved by the different research that utilization of garlic decreases the chances of becoming ill due to the common cold and functioning of the immune system gets better. These positive aspects prove the garlic as a natural immune booster (Percival, 2016).

Anticancer activity

Numerous research results suggested that garlic has potential to express different anti-tumorigenic as well as anticancer activity. Further investigations have proved its potential

due to the presence of different constituents which work as bioactive compounds mainly allyl sulfide and its derivatives. Garlic and it's different preparations have the ability to affect the different steps of cancer genesis such as initiation, proliferation & angiogenesis. However, according to one of the significant research, it has ability to arrest the cell cycle at G2/M phase movement which leads to cell cycle arrest & ultimately responsible for cessation of growth of the cancer cells (Capasso, 2013)

CONCLUSION

Current review discusses the chemical bioactive compound compositions of the Garlic (*Allium sativum*) and its pharmacological properties, which prove the importance of garlic in the cure of different types of diseases. Alliin, Allicin and ajoenes are the primary organo-sulphur bioactive compounds present in the garlic extract. Biological activities of garlic extract have tremendous functional activities which include anti-infectious properties, anti-inflammatory, antioxidant, anti-obesity, cardiovascular protective, anti-diabetic, free radical scavenging and anticancer properties.

ACKNOWLEDGEMENTS

Authors are thankful to School of Life Science & Technology, IIMT University, Meerut.

CONFLICT OF INTEREST

There is no conflict of interest among the authors.

REFERENCES

1. Rasul, S. H. A., Sadiq, B. M. and K. Nauma. (2015). Garlic (Allium sativum): diet based therapy of 21st century review. Asian Pacific Journal of Tropical Disease, 5 (4): 271–278, 2015.
2. Ali, M., Thomson, M. and M. Afzal, M. (2000). Garlic and onions: their effect on eicosanoid metabolism and its clinical relevance. Prostaglandins, Leukotrienes and Essential Fatty Acids, 62: 55–73.
3. Singh, B. B., Vinjamury, S. P. and C. Der-Martirosian, C. (2007). Ayurvedic and collateral herbal treatments for hyperlipidemia: A systematic review of randomized controlled trials and quasi experimental designs. Alternative Therapies in Health and Medicine, 13 (4): 22–28..
4. Lee, D. Y., Li, H., Lim, H. J., Lee, H. J., Jeon, R. and Ryu, J.-H. (2012). Anti-inflammatory activity of sulfur-containing compounds from garlic. *J. Med. Food*, 15, 992–999..
5. Charu, K., Yogita, S. and S. Sonali, S. (2014). Nutraceutical potential of organosulfur compounds in fresh garlic and garlic preparations. Int. J. Pharma Bio Sci., 5: 978-982.
6. Lee, H.S., Lim, W.C., Lee, S.J., Lee, S.H., Lee, J.H. and Cho, H.Y. (2014). Antiobesity effect of garlic extract fermented by lactobacillus plantarum bl2 in diet-induced obese mice. J. Med. Food, 19, 823–829.

7. Bradley, J. M., Organ, C. L. and Lefer, D. J. (2016). Garlic-derived organic polysulfides and myocardial protection. J. Nutr., 2016, 146, 403S–409S.
8. Borek, C. (2006). Garlic reduces dementia and heart-disease risk. Journal of Nutrition, 136 (3): 810–812.
9. Khanum, F., Anilakumar, K. R. and K. R. Viswanathan, K. R. (2004). Anticarcinogenic properties of garlic: a review. Critical Reviews in Food Science and Nutrition, 44: 479–488.
1. Guercio, V., Galeone, C., Turati, C. and Vecchia, L. ((2014). Gastric cancer and allium vegetable intake: a critical review of the experimental and epidemiologic evidence. Nutr. Cancer, 66 (5): 757-773.
1. Lawson., D. K. and Ransom, B.G. (1992). Hughes Inhibition of whole blood platelet-aggregation by compounds in garlic clove extracts and commercial garlic products. Thromb. Res., 65 (2):
1. Lawson L. D. and Hunsaker, S. M. (2018). Allicin Bioavailability and Bioequivalence from Garlic Supplements and Garlic Foods. Nutrients. 10(7):812.
1. Al-Snafi, A. (2013). Pharmacological effects of Allium species grown in Iraq. An overview. Int. J. Pharm. Health Care Res., 1:132–147.
1. Zeng, Y., Li, Y., Yang, J., Pu, X., Du, J., Yang, X., Yang, T. and Yang, S. (2017). Therapeutic Role of Functional Components in Alliums for Preventive Chronic Disease in Human Being. Evid Based Complement Alternat Med. 2017:9402849.
1. Fukushima, S ., Takada, N., Wanibuchi, H., Hori, T., Min, W. and M. Ogawa, M. (2001).Suppression of chemical carcinogenesis by water-soluble organosulfur compounds. J. Nutr., 131: 1049-53.
1. Wallock-Richards, D., Doherty, C. J., Doherty, L., Clarke, D. J., Place, M., Govan, J .R., Campopiano, D. J. (2014). Garlic revisited: Antimicrobial activity of allicin-containing garlic extracts against Burkholderia cepacia complex. PLoS ONE 9, e112726.
1. Kuda, T., Iwai, A. and Yano, T. (2004). E_ect of red pepper Capsicum annuum var. conoides and garlic Allium sativum on plasma lipid levels and cecal microflora in mice fed beef tallow. Food Chem. Toxicol.,, 42: 1695–1700.
1. Ross, Z. M., O'Gara, E. A., Hill, D. J., Sleightholme, H. V. and Maslin, D. J. (2001). Antimicrobial properties of garlic oil against human enteric bacteria: Evaluation of methodologies and comparisons with garlic oil sulfides and garlic powder. Appl. Environ. Microbiol., 67: 475–480.
1. Cutler, R. and Wilson, P. (2004). Antibacterial activity of a new, stable, aqueous extract of allicin against methicillin-resistant Staphylococcus aureus. Br. J. Biomed. Sci. 61: 71–74.
1. Parvu, M., Mot, C.A., Pârvu, A. E., Mircea, C. Stoeber, L., Rosca-Casian, O. and Tigu, A. B. (2019). Allium sativum extract chemical composition, antioxidant activity and antifungal effect against Meyerozyma guilliermondii and Rhodotorula mucilaginosa causing onychomycosis. Molecules, 24: 3958.
1. Fufa, B. 2019. Antibacterial and antifungal properties of garlic extract (Allium sativum): A review. Microbiol. Res. J. Int., 28: 1–5.
1. Gallwitz, H., Bonse, S. and Martinez-Cruz, A. (1999); Schlichting, I.; Schumacher, K.; Krauth-Siegel, R.L. Ajoene is an inhibitor and subversive substrate of human glutathione reductase and Trypanosoma cruzi trypanothione reductase: Crystallographic, kinetic, and spectroscopic studies. J. Med. Chem., 42: 364–372.
1. Abdel-Hafeez, E. H., Ahmad, A. K., Kamal, A. M., Abdellatif, M. Z. and Abdelgelil, N. H. (2015) In vivo anti protozoan effects of garlic (Allium sativum) and ginger (Zingiber officinale) extracts on experimentally infected mice with Blastocystis spp. Parasitol. Res., 114: 3439–3444.

1. Gruhlke, M. C., Nicco, C., Batteux, F. and Slusarenko, A .J. (2016). The effects of allicin, a reactive sulfur species from garlic, on a selection of mammalian cell lines. Antioxidants 6: 1.
1. Ichikawa, M., Yoshida, J., Ide, N., Sasaoka, T., Yamaguchi, H. and Ono, K. (2006). Tetrahydro-β-carboline derivatives in aged garlic extract show antioxidant properties. Journal of Nutrition, 136 (3), Suppl.: 726–731.
1. Liu, J. Z. , Lin, X. Y. and J. A. Milner, J. A. (1992). Dietary garlic powder increases glutathione content and glutathione S-transferase activity in rat liver and mammary tissues.T he FASEB Journal, 6: 3230.
1. Borek, C. (1997). Antioxidants and cancer," Science & Medicine, 4: 51–62.
1. Ide, N. and B. H. S. Lau, B. H. S. (2001). Garlic compounds minimize intracellular oxidative stress and inhibit nuclear factor-βB activation. Journal of Nutrition, 131 (3): 1020–1026.
1. Li, M., Yan, Y. X., Yu, Q. T., Deng, Y., Wu, D. T., Wang, Y., Ge, Y. Z., Li, S. P. and Zhao, J. (2017). Comparison of immunomodulatory effects of fresh garlic and black garlic polysaccharides on RAW 264.7 macrophages.J. Food Sci., 82: 765–771.
1. Liu, T., Zhang, L., Joo, D. et al. (2017). NF-κB signaling in inflammation. Sig Transduct Target Ther 2: 17023.
1. Borek, C. (2001). Antioxidant Health Effects of Aged Garlic Extract .Journal of Nutrition, 131 (3): 1010–1015.
1. A. Tripathi, K. Shrinet, A. Kumar HMGB1 protein as a novel target for cancer Toxicol Rep, 6 (2019), pp. 253-261
1. Percival, S.S. Aged garlic extract modifies human immunity. J. Nutr. 2016, 146, 433S–436S.
1. Capasso A. 2013. Antioxidant action and therapeutic efficacy of Allium sativum L. Molecules, 18: 690-700.